RaceDay — Entity Relationship Diagram (Part 1)

8 entities · 9 relationships · 1:1 extensions for Organiser / Participant · 1:1 Result per Enrolment

1. USERS → ORGANISERS / PARTICIPANTS

1:1 extension tables — Users.Role determines which table the row joins to.

2. ORGANISERS → EVENTS

1:N ownership — each event belongs to exactly one Organiser.

3. EVENTS → CATEGORIES

1:N (e.g. Comrades has Men Open, Women Open, Veterans Men, etc.)

4. PARTICIPANTS + CATEGORY → EVENT ENROLMENTS

N:M join table with UK(ParticipantId, CategoryId); unique per participant+category.

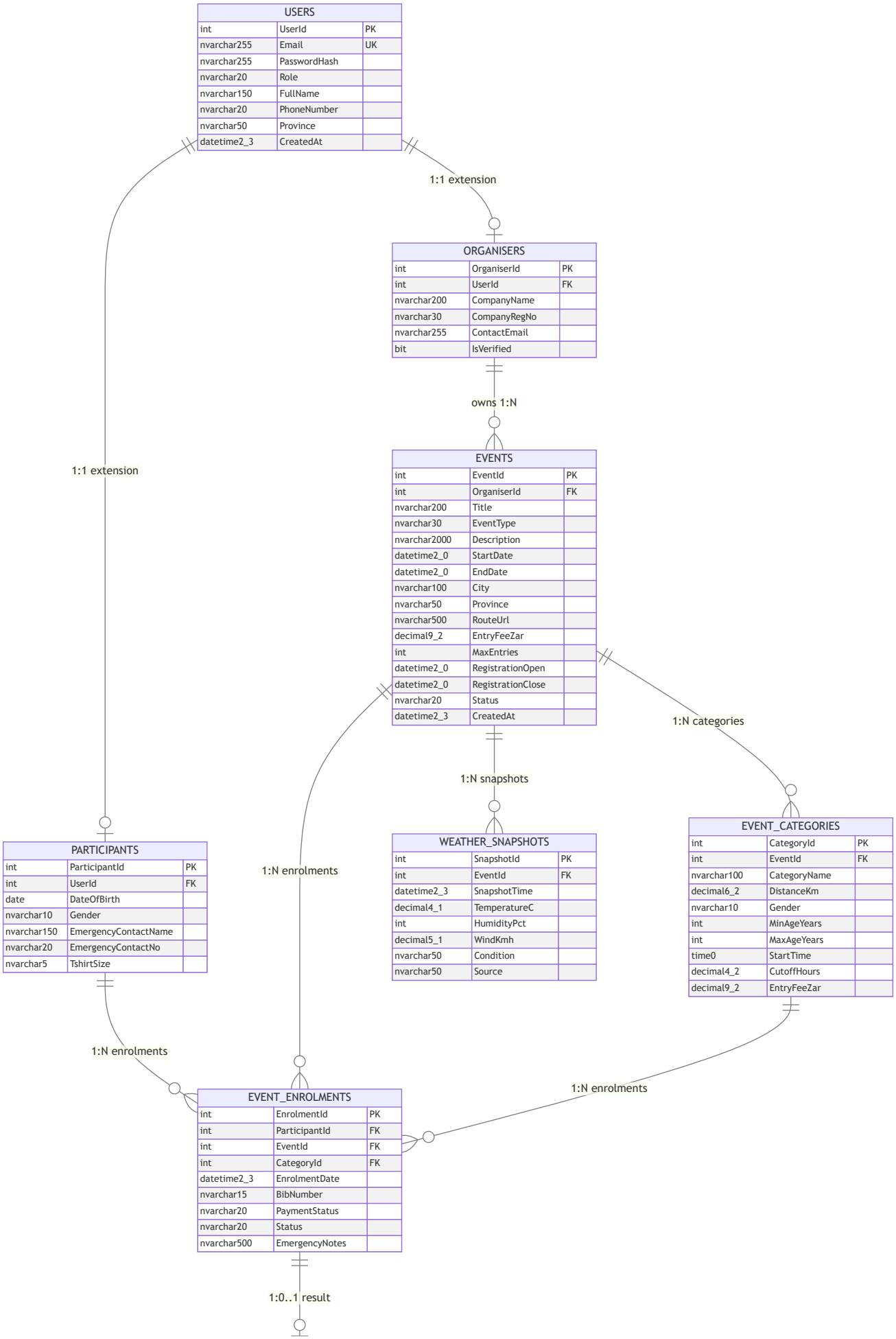
5. ENROLMENTS → RESULTS

1:0..1 — an enrolment produces at most one result (Finished / DNF / DNS / DQ).

6. EVENTS → WEATHER_SNAPSHOTS

1:N — supports "live weather" preparation feature for race day.

Cardinality cheat sheet: `||` = exactly 1, `o|` = 0..1, `}o` = 0..N, `||--o{` = one-to-many (parent required, child optional/many).


USERS
int UserId PK
nvarchar255 Email UK
nvarchar255 PasswordHash
nvarchar20 Role
nvarchar150 FullName
nvarchar20 PhoneNumber
nvarchar50 Province
datetime2_3 CreatedAt
1:1 extension
ORGANISERS
int OrganiserId PK
int UserId FK
nvarchar200 CompanyName
nvarchar30 CompanyRegNo
nvarchar255 ContactEmail
bit IsVerified
owns 1:N
1:1 extension
EVENTS
int EventId PK
int OrganiserId FK
nvarchar200 Title
nvarchar30 EventType
nvarchar2000 Description
datetime2_0 StartDate
datetime2_0 EndDate
nvarchar100 City
nvarchar50 Province
nvarchar500 RouteUrl
decimal9_2 EntryFeeZar
int MaxEntries
datetime2_0 RegistrationOpen
datetime2_0 RegistrationClose
nvarchar20 Status
datetime2_3 CreatedAt
1:N categories
1:N snapshots
1:N enrolments
PARTICIPANTS
int ParticipantId PK
int UserId FK
date DateOfBirth
nvarchar10 Gender
nvarchar150 EmergencyContactName
nvarchar20 EmergencyContactNo
nvarchar5 TshirtSize
WEATHER_SNAPSHOTS
int SnapshotId PK
int EventId FK
datetime2_3 SnapshotTime
decimal4_1 TemperatureC
int HumidityPct
decimal5_1 WindKmh
nvarchar50 Condition
nvarchar50 Source
EVENT_CATEGORIES
int CategoryId PK
int EventId FK
nvarchar100 CategoryName
decimal6_2 DistanceKm
nvarchar10 Gender
int MinAgeYears
int MaxAgeYears
time0 StartTime
decimal4_2 CutoffHours
decimal9_2 EntryFeeZar
1:N enrolments
1:N enrolments
EVENT_ENROLMENTS
int EnrolmentId PK
int ParticipantId FK
int EventId FK
int CategoryId FK
datetime2_3 EnrolmentDate
nvarchar15 BibNumber
nvarchar20 PaymentStatus
nvarchar20 Status
nvarchar500 EmergencyNotes
1:0..1 result

RESULTS		
int	ResultId	PK
int	EnrolmentId	FK
int	FinishTimeSec	
int	ChipTimeSec	
int	GunPosition	
int	CategoryPosition	
int	GenderPosition	
nvarchar20	Status	
nvarchar1000	Splits	
datetime2_3	CapturedAt	

Rendered by Mermaid 10.9.1 — source at `/docs/ERD_MERMAID.md`. The SQL schema in `RaceDay_DB_Script.sql` matches this diagram exactly.